

D606

Data Science Capstone

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

Leaders at a 60-branch credit union tell the capstone team, “We need AI.” Member complaints about loan approval speed rose 35% this year, and the board chair recently attended a fintech conference. What should the first deliverable be?

- A. A problem statement defining slow loan approvals, a measurable target, and the stakeholders affected.
- B. A request for every dataset the credit union has.
- C. A choice of deep learning framework.
- D. A summary of trends from the fintech conference.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

The business requirement at a 300-store pharmacy chain is to reduce out-of-stock events on high-demand medications. The chain’s mobile app was redesigned last year. Which is the best technical requirement?

- A. Produce a daily demand forecast per medication per store, 14 days ahead, with error at or below 15%, feeding the reorder system.
- B. Rebuild the mobile app’s inventory page.
- C. Forecast total chain sales for next year.
- D. Improve inventory management.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A technical requirement reads: classify incoming support tickets into 8 categories within 2 seconds with at least 90% accuracy. Which business requirement most likely produced it?

- A. Predict which customers will cancel.
- B. Route tickets to the right team faster to cut resolution time.
- C. Reduce server hosting costs.
- D. Hire 8 new support teams.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A project to predict equipment failures at a mining company has 14 months of sensor data but only 3 recorded failures. The company operates in two countries. What is the main feasibility concern?

- A. None; any amount of data is enough for machine learning.
- B. Fourteen months is too much data.
- C. Too few failure examples to train a supervised model; consider anomaly detection or gathering more failure data.
- D. Differing regulations in the two countries.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Midway through a 12-week capstone for a dental group, the sponsor asks to add social-media sentiment analysis to a no-show prediction project. The sponsor is enthusiastic and well-liked. What is the best response?

- A. Accept it because the sponsor is well-liked.
- B. Add it and keep the same deadline.
- C. Assess it through change control, prioritize it against the timeline, and likely log it as a later phase.
- D. Refuse all changes once a project starts.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A churn model for a streaming service reaches an AUC of 0.84, and the team celebrates at a rooftop lunch. The business goal was to cut monthly churn from 5% to 4%. How should success be judged?

- A. By the number of customers scored each month.
- B. By the AUC of 0.84 alone.
- C. An AUC above 0.8 always means the project succeeded.
- D. By whether churn actually falls toward 4% against the baseline, with AUC as supporting evidence.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Baseline churn is 5.0% per month among 200,000 subscribers. After a retention program, churn is 4.2%. How many fewer subscribers churn per month?

- A.** 8,400
- B.** 1,600
- C.** 0.8
- D.** 16%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A capstone team at a regional utility with 400,000 meters has finished business understanding and explored the initial data, finding many missing readings. The team works in two-week sprints. Which CRISP-DM phase comes next?

- A.** Modeling
- B.** Data preparation
- C.** Sprint planning
- D.** Evaluation

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

In CRISP-DM evaluation, a model meets its technical metrics but does not meet the business objective. What should happen next?

- A.** Nothing; CRISP-DM phases never repeat.
- B.** Proceed to deployment.
- C.** Rework only the data preparation.
- D.** Return to business understanding to revisit the problem framing and plan.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A hiring-screen model at a 5,000-employee firm excludes gender but uses ZIP code and college name, and it was trained on 10 years of past hiring decisions. HR's offices are being remodeled. What is the main risk?

- A. Proxy variables can reproduce past bias; audit outcomes across groups before use.
- B. Ten years of data is too little.
- C. None, because removing gender guarantees fairness.
- D. Delays from the office remodel.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A project costs \$120,000 to build and \$20,000 a year to run, and it is expected to save \$80,000 a year. The sponsor's department has 14 staff. What is the simple payback period?

- A. 1.5 years
- B. 6 years
- C. 1.75 years
- D. 2 years

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A project must pay back a \$150,000 build cost in 3 years and costs \$30,000 a year to run. What annual savings are needed?

- A. \$30,000
- B. \$20,000
- C. \$80,000
- D. \$50,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

Leaders at a ~~60-branch~~ credit union tell the capstone team, “We need AI.” Member complaints about loan approval speed rose 35% this year, ~~and the board chair recently attended a fintech conference.~~ What should the first deliverable be?

Target word: first deliverable **Decides:** No problem has been defined yet.

A	Correct	Everything else depends on a clear, measurable problem.
B	Adjacent category	Data collection without a question is unfocused.
C	Inverted operation	Tool before problem.
D	Camouflage	The chair’s conference does not define the problem.

2. Answer A Concept 2

The business requirement at a ~~300-store~~ pharmacy chain is to reduce out-of-stock events on high-demand medications. ~~The chain’s mobile app was redesigned last year.~~ Which is the best technical requirement?

Target word: best technical translation **Decides:** Needs output, detail level, timing, and a quality bar.

A	Correct	Specific output, granularity, horizon, and measurable quality.
B	Camouflage	The app redesign does not address stockouts.
C	Adjacent category	Wrong level of detail for store-level stockouts.
D	Dropped element	Restates the goal with nothing measurable.

3. Answer B Concept 2 · transfer

A technical requirement reads: classify incoming support tickets into 8 categories within 2 seconds with at least 90% accuracy. Which business requirement most likely produced it?

Target word: reverse: technical back to business **Decides:** Classifying tickets on arrival supports routing.

A	Confusion set	A churn goal would need a churn prediction.
B	Correct	Fast, accurate categorization serves routing.
C	Adjacent category	Nothing in the spec targets cost.
D	Wrong stem number	Echoes the 8 categories; not a data requirement.

4. Answer C Concept 3

A project to predict equipment failures at a mining company has 14 months of sensor data but only 3 recorded failures. ~~The company operates in two countries.~~ What is the main feasibility concern?

Target word: feasibility risk **Decides:** Only 3 examples of the outcome.

A	Absolute language	Models need enough examples of each outcome.
B	Inverted operation	The shortage is in failures, not in time span.
C	Correct	A classifier cannot learn a class from 3 cases.
D	Camouflage	Does not address whether the data supports the model.

5. Answer C Concept 4

Midway through a 12-week capstone for a dental group, the sponsor asks to add social-media sentiment analysis to a no-show prediction project. ~~The sponsor is enthusiastic and well liked.~~ What is the best response?

Target word: handling a scope change **Decides:** A new feature mid-project with a fixed timeline.

A	Camouflage	Likability does not decide scope.
B	Dropped element	Ignores the time and resource impact.
C	Correct	Protects the core deliverable while taking the request seriously.
D	Absolute language	Some changes are worth making; they need evaluation.

6. Answer D Concept 5

A churn model for a streaming service reaches an AUC of 0.84, ~~and the team celebrates at a rooftop lunch.~~ The business goal was to cut monthly churn from 5% to 4%. How should success be judged?

Target word: success criterion **Decides:** The goal is a churn rate, not a model score.

A	Confusion set	Measures activity, not outcome.
B	Adjacent category	A good model may not change churn.
C	Absolute language	No model metric guarantees business impact.
D	Correct	The business KPI defines success.

7. Answer B Concept 5 · transfer

Baseline churn is 5.0% per month among 200,000 subscribers. After a retention program, churn is 4.2%. How many fewer subscribers churn per month?

Target word: business impact in subscribers **Decides:** 0.8 percentage points of 200,000.

A	Wrong stem number	Subscribers who still churn.
B	Correct	$10,000 - 8,400 = 1,600$.
C	Dropped element	Percentage points without applying them to the base.
D	Adjacent category	The relative reduction, not a subscriber count.

8. Answer B Concept 6

A capstone team at a regional utility ~~with 400,000 meters~~ has finished business understanding and explored the initial data, finding many missing readings. ~~The team works in two-week sprints.~~ Which CRISP-DM phase comes next?

Target word: next phase **Decides:** Data understanding is done; quality issues found.

A	Dropped element	Skips fixing the data problems.
B	Correct	Cleaning and handling the missing readings comes before modeling.
C	Camouflage	A work cadence, not a CRISP-DM phase.
D	Confusion set	Evaluation judges a built model.

9. Answer D Concept 6 · transfer

In CRISP-DM evaluation, a model meets its technical metrics but does not meet the business objective. What should happen next?

Target word: where to loop back **Decides:** The business objective was not met.

A	Absolute language	The model is explicitly iterative.
B	Dropped element	Ignores the business failure.
C	Adjacent category	The gap is in the business fit, not just the data.
D	Correct	CRISP-DM loops back when evaluation fails the business goal.

10. Answer A Concept 7

A hiring-screen model at a 5,000-employee firm excludes gender but uses ZIP code and college name, and it was trained on 10 years of past hiring decisions. ~~HR's offices are being remodeled.~~ What is the main risk?

Target word: fairness risk **Decides:** Proxies plus biased historical labels.

A	Correct	Correlated fields can stand in for the removed attribute.
B	Dropped element	The issue is what the data encodes, not its length.
C	Absolute language	Proxies can carry the same signal.
D	Camouflage	Unrelated to model risk.

11. Answer D Concept 8

A project costs \$120,000 to build and \$20,000 a year to run, and it is expected to save \$80,000 a year. ~~The sponsor's department has 14 staff.~~ What is the simple payback period?

Target word: payback period **Decides:** Net annual benefit = \$60,000.

A	Dropped element	Ignores the running cost: $120 / 80$.
B	Wrong stem number	Divides build cost by running cost.
C	Confusion set	Treats the running cost as one-time: $140 / 80$.
D	Correct	$\$120,000 / (\$80,000 - \$20,000) = 2$.

12. Answer C Concept 8 · transfer

A project must pay back a \$150,000 build cost in 3 years and costs \$30,000 a year to run. What annual savings are needed?

Target word: solve for required savings **Decides:** Net benefit needed = \$50,000 per year.

A	Wrong stem number	The running cost from the stem.
B	Inverted operation	Subtracts running cost instead of adding it.
C	Correct	$\$150,000 / 3 = \$50,000$ net + \$30,000 running = \$80,000.
D	Dropped element	Forgets to cover the running cost.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	3	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Start with a problem statement	A capstone begins with a measurable business problem, its stakeholders, and the decision it supports, not with a technology.	Starting from a tool or technique (“we need AI”).	A leader asks for a technology with no defined problem.
2	Business to technical requirements	Translate a business need into a technical requirement that names the output, its level of detail, timing, and a measurable quality bar.	Restating the business goal vaguely or at the wrong level of detail.	A business goal next to candidate technical specs.
3	Data feasibility	Check that data exists in the right quantity and quality, especially enough examples of the outcome, before committing to a method.	Assuming any amount of data supports supervised learning.	Very few positive cases, or a key field missing.
4	Scope control	Handle new requests through change control and prioritization (such as MoSCoW) against time and resources.	Accepting every addition, or refusing every change.	A mid-project request to add a new feature.
5	Success criteria	Judge success by the business KPI against a baseline; model metrics support but do not replace it.	Declaring success because a model metric looks good.	A model metric reported next to a business goal.
6	CRISP-DM phases	Business understanding → data understanding → data preparation → modeling → evaluation → deployment, looping back when evaluation fails the business goal.	Skipping preparation, or treating the phases as one-way.	“Just finished” one phase; what comes next?
7	Bias and proxy variables	Removing a protected attribute does not ensure fairness; correlated proxies can reproduce historical bias, so audit outcomes across groups.	Believing that dropping gender or race makes a model fair.	A model trained on past decisions using ZIP code or similar fields.
8	Cost-benefit and payback	Simple payback = up-front cost / (annual savings – annual running cost).	Ignoring running costs or treating them as one-time.	Build cost, run cost, and annual savings.

Course scope (public description): Data Science Capstone integrates the key concepts from the MSDA core with the knowledge gained in the three courses within the Data Science specialization. In this course, students will evaluate various needs and opportunities in an organization or marketplace. In addition, students will identify business requirements and translate those business requirements into technical requirements. Finally, students will create a comprehensive project plan to solve a problem in a way that satisfies customer or business needs. Projects within this specialization may include the design and construction of machine learning approaches, optimization, and the use of advanced analytics techniques as the project requires. There is no prerequisite for this course.